

INMO(30th) – 2015

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be a right-angled triangle with $\angle B = 90^\circ$. Let BD be the altitude from B on to AC. Let P, Q and I be the incentres of triangles ABD, CBD and ABC respectively. Show that the circumcentre of the triangle PIQ lies on the hypotenuse AC.
 2. For any natural number $n > 1$, write the infinite decimal expansion $\frac{1}{n}$ (for example, we write $\frac{1}{2} = 0.4\overline{9}$ as its infinite decimal expansion, not 0.5). Determine the length of the non-periodic part of the (infinite) decimal expansion of $\frac{1}{n}$.
 3. Find all real function $f : R \rightarrow R$ satisfying the relation: $f(x^2 + yf(x)) = xf(x + y)$.
 4. There are four basket-ball players A, B, C, D. Initially, the ball is with A. The ball is always passed from one person to a different person. In how many ways can the ball come back to A after seven passes? (For example $A \rightarrow C \rightarrow B \rightarrow D \rightarrow A \rightarrow B \rightarrow C \rightarrow A$ and $A \rightarrow D \rightarrow A \rightarrow D \rightarrow C \rightarrow A \rightarrow B \rightarrow A$ are two ways in which the ball can come back to A after seven passes.)
 5. Let ABCD be a convex quadrilateral. Let the diagonals AC and BD intersect in P. Let PE, PF, PG and PH be the altitudes from P on to the sides AB, BC, CD and DA respectively. Show that ABCD has an incircle if and only if $\frac{1}{PE} + \frac{1}{PG} = \frac{1}{PF} + \frac{1}{PH}$.
 6. Show that from a set of 11 square integers one can select six numbers $a^2, b^2, c^2, d^2, e^2, f^2$ such that $a^2 + b^2 + c^2 = d^2 + e^2 + f^2 \pmod{12}$.

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